

Deep Learning - Theory and Practice

Basics of Machine Learning

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<http://leap.ee.iisc.ac.in/sriram/teaching/DL19/>

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Problems in Decision Theory

- ❖ Decision Theory
 - ❖ Inference problem
 - ❖ Finding the joint density $p(\mathbf{x}, \mathbf{t})$
 - ❖ Decision problem
 - ❖ Using the inference to make the classification or regression decision

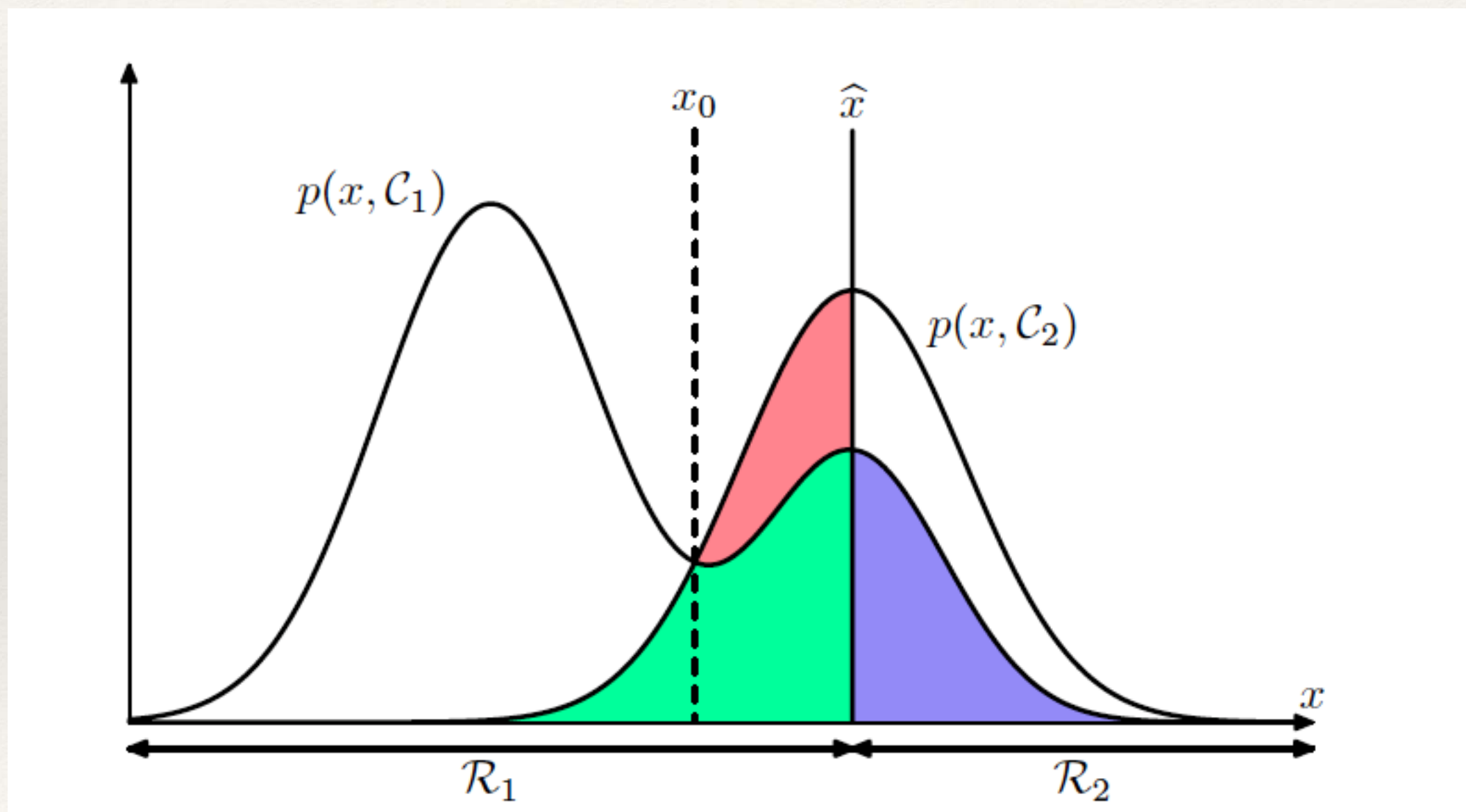
Decision Problem - Classification

- ❖ Minimizing the mis-classification error
- ❖ Decision based on maximum posteriors

$$\mathit{argmax}_j p(C_j|\mathbf{x})$$

- ❖ Loss matrix
 - ❖ Can be used for non uniform error weighting.

Decision Theory



Approaches for Inference and Decision

I. Finding the joint density from the data.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

II. Finding the posteriors directly.

Neural Networks

III. Using discriminant functions for classification.

Advantages of Posteriors

- ❖ Minimizing the risk
- ❖ Reject Option
- ❖ Combining models
- ❖ Compensating for class priors

Loss Function for Regression

- ❖ With a mean square error loss

$$\mathbb{E}[L] = \iint \{y(\mathbf{x}) - t\}^2 p(\mathbf{x}, t) \, d\mathbf{x} \, dt.$$

- ❖ The problem boils down to conditional expectation of the data given the

Regression Problem

